

- Functional Requirements Document (FRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Document Version:** 1.0
- Prepared by:** [Your Name], Senior Business Analyst
- Date:** [Current Date]

1. Introduction

This Functional Requirements Document (FRD) outlines the functional requirements for the data integration and aggregation project involving customer and order data.

2. Functional Requirements

The following functional requirements have been identified:

1. **Data Loading**:

- Load customer data from CSV file
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into `customer` delta table.
- Load order data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into `order` delta table.

2. **Data Cleansing**:

- Remove null records from `customer` and `order` tables.
- Remove duplicate records from `customer` and `order` tables.

3. **Data Transformation**:

- Join `customer` and `order` tables using `CustId` field.
- Create `ordersummary` table with schema: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.

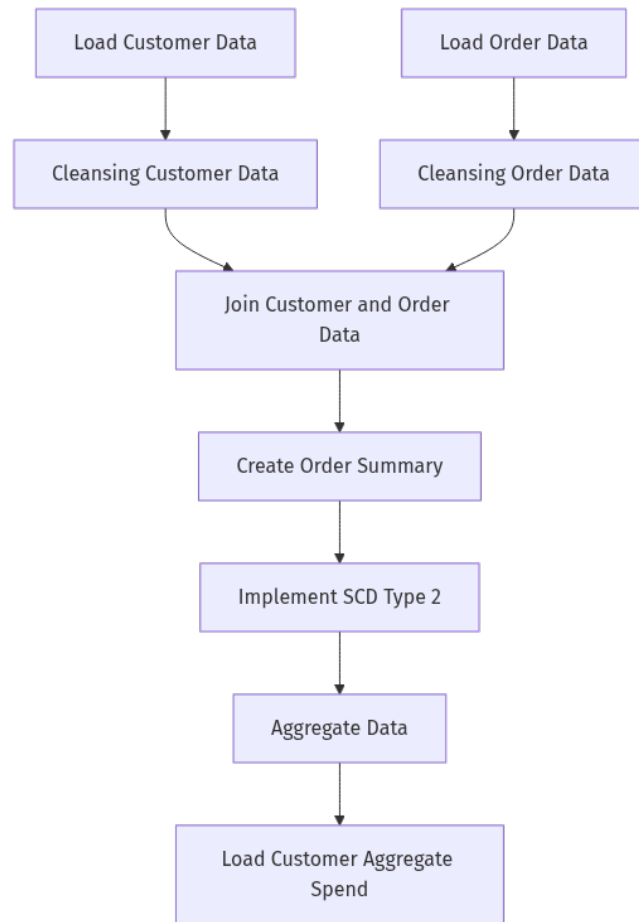
4. **SCD Type 2 Implementation**:

- Implement SCD type 2 logic on `ordersummary` table.
- Update `StartDate` and `EndDate` columns accordingly.

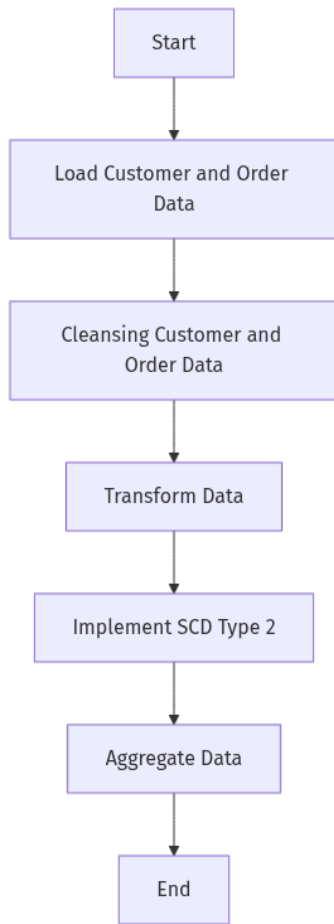
5. **Data Aggregation**:

- Aggregate data from `ordersummary` table.
- Create `customeraggregatespend` table with schema: `Name`, `TotalAmount`, `Date`.

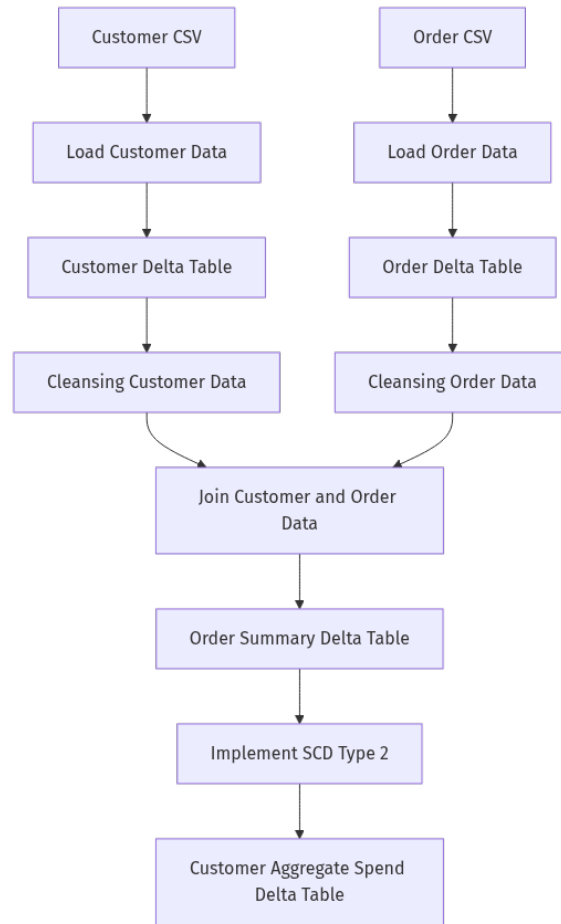
3. System Flow Diagram



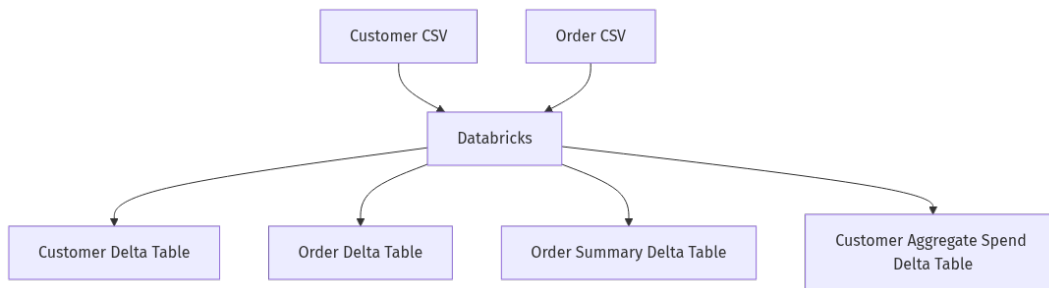
4. Process Flow / Flowchart



5. Data Flow Diagram (DFD)



6. High-Level Architecture Diagram



The above FRD provides a concise overview of the functional requirements for the data integration and aggregation project. The included diagrams illustrate the system flow, process flow, data flow, and high-level architecture of the proposed solution.